Depeg Tracker + DEWS Changelog

Full version history of Depeg Tracker and DEWS methodology decisions, from v1.0 to v6.0.

ump to Version

Latest v6.0 v5.99 v5.98 v5.97 v5.96 v5.95 v5.94 v5.93 v5.92 v5.91 v5.9

LATEST VERSION

v6.0 May 14, 2026

Historical provenance, quality-aware PegScore, and calibration metrics

Depeg history now records durable replay/audit provenance, PegScore discounts weak or disputed historical rows, and DEWS backtests report calibration metrics from stored stress-signal history.

IMPACT SNAPSHOT

- Backfill runs persist replay status, counts, fingerprints, source providers, quote mode, peg-reference/supply source, confidence tier, and public provenance projections
- Audit verdicts are direction-aware and persist confirmed, disputed, false_positive, no_data, and repaired-style outcomes without deleting rows solely because audit data is weak or absent
- PegScore excludes false-positive/disputed events and downweights low-confidence rows while surfacing quality-adjustment counters to callers
- DEWS calibration metrics now use stored stress_signal_history as the primary source and report precision, recall, false-positive days, false-negative incidents, lead-time percentiles, churn, and cohorts

v6.0 May 14, 2026

Historical provenance, quality-aware PegScore, and calibration metrics

Depeg history now records durable replay/audit provenance, PegScore discounts weak or disputed historical rows, and DEWS backtests report calibration metrics from stored stress-signal history.

- Backfill runs persist replay status, counts, fingerprints, source providers, quote mode, peg-reference/supply source, confidence tier, and public provenance projections
- PegScore excludes false-positive/disputed events and downweights low-confidence rows while surfacing quality-adjustment counters to callers
- Audit verdicts are direction-aware and persist confirmed, disputed, false_positive, no_data, and repaired-style outcomes without deleting rows solely because audit data is weak or absent
- DEWS calibration metrics now use stored stress_signal_history as the primary source and report precision, recall, false-positive days, false-negative incidents, lead-time percentiles, churn, and cohorts

Hide details

IMPACT NOTES

- Backfill runs persist replay status, counts, fingerprints, source providers, quote mode, peg-reference/supply source, confidence tier, and public provenance projections
- PegScore excludes false-positive/disputed events and downweights low-confidence rows while surfacing quality-adjustment counters to callers
- Audit verdicts are direction-aware and persist confirmed, disputed, false_positive, no_data, and repaired-style outcomes without deleting rows solely because audit data is weak or absent
- DEWS calibration metrics now use stored stress_signal_history as the primary source and report precision, recall, false-positive days, false-negative incidents, lead-time percentiles, churn, and cohorts

Commit provenance not recorded

v5.99 May 13, 2026

Structured Yield Source-Risk DEWS Input

DEWS now consumes populated Yield Intelligence source-risk and rank-attribution fields inside the existing Yield Anomaly sub-signal, while neutral or missing structured rows remain no-ops.

- The `yield` sub-signal can now score populated `sourceRisk.rewardShare`, `sourceRisk.sourceDepthRatio`, `sourceRisk.sourceAgeSeconds`, `sourceRisk.sourceSwitchCount30d`, `sourceRisk.sourceRiskPenalty`, high reviewed `venueRiskTier`, and rank-attribution source-risk/source-switch drivers
- Structured yield evidence is additive with existing `yield_data.warning_signals` and still caps the Yield Anomaly sub-signal at 100

- Neutral structured rows do not become available zero-stress signals, so missing or neutral evidence does not dilute the DEWS weighted average
- The DEWS source-state loader reads structured evidence from the published `yield-rankings` cache and preserves legacy warning-only behavior when that cache is missing or malformed

Details

v5.98 May 12, 2026

Registry-backed depeg source families

Depeg trust and pending-confirmation policy now resolve source families from pricing-source registry metadata, expanding composite labels before family and authority checks.

- Composite labels such as `coingecko+geckoterminal` are expanded before source-family checks, so they no longer behave like unknown standalone sources
- CoinGecko variants, DefiLlama list/detail/contract variants, and CoinMarketCap-style list aggregators remain correlated families for confirmation and severe-downside corroboration
- Fallback/search lanes remain non-authoritative, while hard market, oracle, protocol, and promoted DEX protocol lanes keep explicit registry-backed families

Details

v5.97 May 11, 2026

Source-family confirmation and zero-event backfill parity

Pending depeg confirmation now requires source-family-aware corroboration, stricter DEX/pool independence, refreshed peg references, and canonical confirmer provenance; trusted zero-event backfills now remove stale backfill rows.

- CoinGecko/DefiLlama off-chain confirmation is selected from the primary agreeSources family set, so same-family circular evidence cannot promote large-cap pending rows on its own
- Aggregate DEX confirmation requires at least two fresh protocol groups, and pool challengers count distinct protocol/source-family groups while preserving the documented >= $5M single-pool exception
- Promoted pending rows use refreshed peg references, canonical confirmation_sources keys, and the worst trustworthy confirmer price when setting peak severity
- Backfill dry-runs and mutating runs now agree when a trusted replay finds zero events: stale source='backfill' rows are previewed and then deleted

Details

v5.96 May 5, 2026

ARS native-peg and KGS FX corroboration

Direct native-peg corroboration now includes ARS where CoinGecko exposes a supported quote currency, while KGS stays on the secondary daily FX path and deterministic bounds because CoinGecko does not expose a native KGS quote.

- WARS (ARS) live and pending depeg checks can use fresh native CoinGecko quotes before trusting USD/FX-derived divergence on its own
- Historical replay behavior is unchanged except that assets with native ARS market history can prefer that direct native series when available
- KGST (KGS) depeg checks use the secondary daily FX mirror and hardcoded validation bounds rather than an unsupported CoinGecko KGS quote

Details

v5.95 Apr 18, 2026

Cross-asset contagion amplifier

DEWS now applies a bounded per-peg-type contagion amplifier (max 1.2x) derived from the same cycle's first-pass DANGER/WARNING bands, on top of the existing systemic PSI amplifier.

- A tracked stablecoin entering DANGER/WARNING now raises other same-peg-type coins by up to 15% under current constants (1.15x / 1.08x); the amplifier is defensively capped at 20%
- Amplifier is clamped, explainable (no learned weights), and surfaced on /api/stress-signals as amplifiers.contagion
- First-pass coins that themselves are DANGER/WARNING do not contagion-amplify themselves

Details

v5.94 Apr 18, 2026

Pool-confirmation hardening, backfill atomicity, confirmation-provenance surfacing

Pool-only pending promotion now requires 2 pools or >= $5M TVL; backfill delete+insert share a batch; off-chain confirmation is circuit-breaker-guarded; promoted depeg events now persist confirmation_sources and pending_reason.

- Single-pool manipulation can no longer unilaterally promote a pending depeg (bar = 2
- A worker interruption during backfill no longer leaves a coin with zero depeg rows

pools OR one pool with >= $5M TVL)

- A CoinGecko/DefiLlama outage no longer hammers the endpoint for 45 min per pending row
- Promoted events carry confirmation_sources (e.g. 'DEX+CEX') and pending_reason (e.g. 'large-cap+low-confidence') for ex-post diagnostics

Details

v5.93 Apr 15, 2026

Blacklist signal coverage follows direct EVM wave

DEWS blacklist-activity input now follows the direct EVM blacklist tracker expansion, adding supported event-count stress signals for FDUSD, BRZ, AUSD, MNEE, EURI, USDQ, USDO, USDX, AID, TGBP, EURC, and BUIDL where those assets are otherwise DEWS-eligible.

- The shared supported blacklist symbol set now includes the direct EVM coverage wave
- Non-USD amount valuation remains owned by the blacklist tracker; DEWS uses event counts only
- DEWS remains coupled to that shared live tracker set instead of maintaining a separate symbol list

Details

v5.92 Apr 15, 2026

Blacklist signal coverage follows first-wave tracker expansion

DEWS blacklist-activity input now follows the expanded live blacklist tracker symbol set, adding first-wave issuer-intervention signals for USDG, RLUSD, U, USDtb, and A7A5 where those assets are otherwise DEWS-eligible.

- The shared supported blacklist symbol set now includes USDG, RLUSD, U, USDTB, and A7A5
- A7A5's blacklist signal can participate as event-count stress only; non-USD amount valuation remains owned by the blacklist tracker ledger
- DEWS continues to derive blacklist-signal eligibility from that shared set instead of maintaining a separate list

Details

v5.91 Apr 11, 2026

Conservative DEWS freshness and zero-supply current-row retirement

DEWS aggregate responses now expose oldest-row lag separately, and eligible assets with no current circulating supply no longer keep stale current radar rows.

- The aggregate `/api/stress-signals` response preserves `updatedAt` as the newest row used for freshness headers while adding `oldestComputedAt` as a body-only lag diagnostic
- Last-valid cached rows still remain available for coins that have positive current supply but insufficient signal coverage in an individual cycle
- The DEWS cron now retires current `stress_signals` rows for PSI-eligible assets that are explicitly present in the stablecoins cache with zero current supply, without deleting daily history

Details

v5.9 Apr 8, 2026

Direction-true confirmation, pending refreshes, and DEWS live-trust alignment

Pending depeg incidents now track live first/last/peak state, confirmation requires same-direction corroboration, and DEWS divergence reuses the live depeg DEX trust floor.

- Pending rows are now refreshed while they wait for confirmation, preserving first-seen timestamps but updating last-seen and worst-seen state for the active direction
- DEWS divergence now ignores fresh-but-thin DEX rows unless they pass the same `$1M` live depeg trust gate, while the repair path refreshes current rows and prunes unrecomputable daily history from the Mar 9, 2026 trust-floor boundary onward
- Pending confirmation now treats opposite-side secondary evidence as contradiction instead of support, and native-quote recovery no longer persists contradictory `recovery_price` values

Details

v5.8 Apr 7, 2026

Daily-confirmed native-peg historical replay

Supported non-USD fiat historical replay now treats native-fiat CoinGecko history as a day-scale confirmation lane instead of trusting thin hourly native prints on their own.

- Historical native-fiat replay now uses daily points plus a two-point confirmation window across 36 hours before opening normal non-USD fiat backfill events
- Broad non-USD backfill repairs can now use authenticated CoinGecko market-chart transport consistently through the admin/backfill path, reducing false rebuild drift during large historical cleanups
- Extreme single-point native crashes of 5,000 bps or more are still preserved even when the native historical confirmation rule would otherwise suppress them

Details

v5.7 Apr 7, 2026

Launch-date peg-score anchors for older tracked assets

PegScore tracking windows now prefer curated launch dates over late-arriving supply-history coverage when the asset metadata provides a trustworthy launch anchor.

- PegScore now uses a curated launch date as the age anchor when one is available, falling back to earliest supply-history coverage only when no launch date is curated
- BRZ now anchors peg-score tracking to its July 19, 2019 launch date instead of a March 2025 supply-history coverage artifact
- Older tracked assets with late `supply_history` coverage no longer appear artificially young in the peg-score window

Details

v5.6 Apr 7, 2026

Generalized native-peg routing and replay for non-USD fiat assets

Live and pending depeg routing now generalize the native-peg corroboration lane across the supported non-USD fiat set, and historical replay prefers native fiat market history where CoinGecko exposes it.

- Fresh direct native-peg quotes can now veto, sustain, or resolve live depeg state for supported EUR/CHF/GBP/JPY/SGD/AUD/CAD/BRL/IDR/TRY/ZAR/PHP/MXN/RUB/CNH-style fiat pegs instead of remaining effectively BRL-only
- Historical backfill now replays supported non-USD fiat assets against direct native fiat price history and a native `1.0` peg when that series exists, which removes large classes of synthetic backfill events caused by USD/FX mismatch
- Pending confirmation prefers that same direct native quote first whenever CoinGecko exposes the matching fiat pair, reducing false confirmations from USD/FX reference drift

Details

v5.5 Apr 7, 2026

Direct native-peg corroboration for BRL depeg routing

Live and pending depeg routing now checks a fresh direct native-peg quote for supported non-USD fiat assets before trusting a USD-price-versus-FX-reference divergence on its own.

- BRZ-style depegs can now be vetoed or resolved by a fresh direct BRL quote when the USD/FX-derived signal is back inside threshold or points the other way
- Prevents thin BRL reference mismatches from opening, sustaining, or confirming false depeg rows while still preserving genuine native-peg stress
- Pending confirmation prefers the fresh direct native-peg quote over a weaker derived USD cross-check when CoinGecko exposes that native pair

Details

v5.4 Apr 7, 2026

Thin-fiat peg-reference fail-closed and corroborated primary recovery

Live depeg state now fails closed when thin non-USD fiat peg references lose their FX fallback, while fresh multi-source primary agreement can retire stale live rows once the coin is back inside threshold.

- Thin fiat peg groups with fewer than 3 live contributors now require cached FX fallback before they can open, update, or resolve live depeg state
- Prevents BRL-style peer-median false positives from both opening new live rows and lingering as active depegs after the peg reference normalizes
- Fresh non-cached multi-source primary agreement can now close an already-open live event after recovery even if that source mix is still too soft to open new events directly

Details

v5.3 Apr 6, 2026

DEWS flow baseline continuity on quiet 24-hour windows

DEWS no longer drops the mint/burn flow signal just because the latest 24-hour window had zero activity while a mature 30-day baseline still exists.

- Coins with >= 7 days of mint/burn history now keep the flow signal available even when the latest 24-hour bucket is quiet
- DEWS cron assembly now reflects the documented mint/burn-baseline semantics for mature tracked coins
- A zero-flow day now contributes zero flow stress instead of silently redistributing the flow weight away from the score

Details

v5.2 Apr 3, 2026

Corroborated DEX recovery gating for live depeg state

Aggregate DEX bridge rows no longer count as sufficient evidence on their own for ambiguous-primary recoveries or recovery-style event suppression.

- DEX-assisted live recovery now requires at least two corroborating protocol-level DEX groups inside threshold, not just one trusted aggregate row
- New-event suppression and other aggregate-DEX-driven state mutations now share that stricter corroboration policy, reducing synthetic split events on chronically depegged coins
- Large challenger pools can veto ambiguous-primary DEX recoveries when they still show the old depeg direction

Details

v5.1 Mar 31, 2026

Ongoing depeg continuity over DEX-only contradiction

Already-open depeg events no longer auto-close just because a trusted aggregate DEX row temporarily moves back inside the threshold while the primary path still shows the same-direction depeg.

- Same-direction DEX disagreement is now advisory for ongoing events instead of forcing a synthetic recovery boundary
- Prevents repeated event fragmentation on chronically depegged assets when aggregate DEX pricing briefly snaps back toward peg
- Open depeg rows remain continuous until the standard recovery path confirms a sub-threshold resolution

Details

v5.0 Mar 28, 2026

DEWS blacklist coverage parity and thin-peg FX fallback parity

DEWS now applies issuer-freeze signal coverage to the full live blacklistable set and derives thin non-USD peg references with the same cached FX fallback path used elsewhere in the peg-aware pipeline.

- Blacklist signal coverage is now derived from the shared supported blacklist symbol set, so PYUSD and USD1 receive the same DEWS blacklist input treatment as USDC, USDT, PAXG, and XAUT
- DEWS input semantics now match the documented blacklist coverage and the live depeg / peg-summary FX-reference path
- Thin non-USD DEWS divergence inputs now use cached `fxFallbackRates` from the stablecoins payload instead of relying only on sparse peer medians

Details

v4.9 Mar 23, 2026

Bootstrap sentinel and core-liquidity freshness gating

DEWS bootstrap is now a one-time state transition, and stale or missing core liquidity inputs no longer masquerade as acceptable startup conditions.

- Bootstrap grace now ends after the first successful DEWS publication via a dedicated `dews:bootstrap-complete` sentinel instead of piggybacking on stablecoins-cache freshness
- Fresh `dex_liquidity` is now required for publication, with rows older than 2 hours treated as a hard degraded source failure
- Only explicitly optional missing tables remain bootstrap-allowed before first success; core dependencies no longer inherit that grace

Details

v4.8 Mar 22, 2026

Contradicted live depegs now retire into pending confirmation

When a low-confidence primary price now contradicts an open live depeg across the peg, the stale live row is retired immediately and the replacement move waits in pending confirmation instead of leaving the wrong direction active.

- Opposite-direction live depeg rows no longer remain active just because the correcting primary price is still confirm_required
- Public active-depeg state stops claiming the wrong side of the peg while confirmation catches up on the corrected move
- Direction flips from cached, fallback, low-confidence, or stale primary inputs now close the stale live row and insert a replacement pending candidate

Details

v4.7 Mar 21, 2026

Early peg score: minimum data threshold lowered from 30 to 7 days

Peg score is now emitted after 7 days of tracking instead of 30, with an 'Early score' label for the 7–30 day window.

- Minimum tracking threshold reduced from 30 to 7 days — coins receive a composite peg score after
- Scores based on 7–30 days of data are labelled 'Early score' in the hero card (amber text with

their first week

tooltip)

- Report card peg-stability dimension is now rated from day 7; NR only appears for coins with < 7 days of history

Details

v4.6 Mar 10, 2026

Confidence-aware depeg routing, extreme-move confirmation, and provenance surfacing

Depeg detection stopped treating every non-null price as equally trustworthy and now routes ambiguous or catastrophic moves through explicit confirmation paths.

- Cached, fallback, low-confidence, and stale primary prices now require confirmation before they can open or close live depeg state
- peg-summary now exposes price provenance and trust state, and the depeg page consumes backend freshness metadata plus real event-history pagination
- Extreme moves no longer get dropped just for crossing the old <0.5x or >2x peg guardrail; they enter a dedicated confirmation lane instead

Details

v4.5 Mar 9, 2026

Trusted DEX-price gating for depeg suppression, confirmation, and UI checks

DEX cross-validation now shares an explicit trust policy so thin pools cannot suppress or confirm depegs, and low-confidence rows no longer surface on the public DEX Price Check UI.

- Depeg suppression/confirmation now requires fresh DEX rows with >= $1M aggregate source TVL
- Thin DEX rows can no longer veto new depeg events or promote pending confirmations on their own
- UI-facing dexPriceCheck exposure now requires fresh data with >= $250K aggregate source TVL

Details

v4.4 Mar 2, 2026 Reconstructed

No-history coins now return null peg score

Peg score stopped treating coins with neither first-seen supply history nor depeg events as implicitly healthy.

- coinTrackingStart now returns null when both firstSeen and events are absent
- Prevents false perfect-score outcomes on sparse or incomplete datasets
- computePegScoreWithWindow now yields null pegScore for insufficient-history coins

Details

v4.3 Mar 1, 2026 Reconstructed

Young-coin fairness and stronger active penalties

Peg score became less permissive for young coins with recurring brief depegs and for currently depegged assets.

- Tracking start formalized as max(firstSeen, fourYearsAgo) with earliest-event fallback
- Active-depeg penalty steepened to max(5, absBps/50), capped at 50
- Per-event severity now applies max(duration penalty, magnitude floor)

Details

v4.2 Mar 1, 2026 Reconstructed

DEWS wave-2: yield signal + PSI amplifier

DEWS expanded beyond market microstructure signals by incorporating yield-warning telemetry and systemic PSI context.

- Added 8th DEWS sub-signal: yield anomaly (weight 0.05)
- Cron now reads yield_data.warning_signals and latest PSI sample before scoring
- Introduced systemic amplifier: DEWS boosted up to +30% when PSI < 75

Details

v4.1 Mar 1, 2026 Reconstructed

DEWS pool stress calibration fix

Corrected pool-stress scaling error that was inflating DEWS pool signal values.

- avg_pool_stress is now consumed as native 0-100 (removed erroneous x100)
- Reduced false high-band classifications caused by scale inflation
- Pool component returned to intended weighting and magnitude

Details

v4.0 Mar 1, 2026 Reconstructed

DEWS launch (7-signal model + 15-minute cron)

Launched the Depeg Early Warning System with per-coin stress scoring and persisted 15-minute snapshots.

- Introduced DEWS base model with 7 sub-signals and weighted-redistribution scoring
- compute-dews cron writes rolling stress_signals and daily stress_signal_history snapshots
- Threat bands established: CALM, WATCH, ALERT, WARNING, DANGER

Details

v3.2 Feb 27, 2026 Reconstructed

Tracking-window direction fix

Corrected tracking-start math to avoid diluting young-coin depeg severity across pre-launch periods.

- Lookback boundary corrected from min(firstSeen, fourYearsAgo) to max(...)
- Young-coin scores became less artificially inflated
- Peg-time and severity now computed against realistic coin lifetime bounds

Details

v3.1 Feb 26, 2026 Reconstructed

Confirmation and detector hardening

Hardened confirmation and detection against invalid references, non-finite values, and partial-write edge cases.

- Pending rows with invalid peg_reference are dropped before confirmation math
- Pending confirmation mutations are batched atomically for consistent state transitions
- Detection now rejects non-finite peg references before bps computation

Details

v3.0 Feb 25, 2026 Reconstructed

Two-stage confirmation for large-cap depegs

Large-cap depeg detection moved from single-source trigger to a pending-confirmation pipeline with secondary-source validation.

- >= $1B coins now route through depeg_pending before promotion to depeg_events
- sync-stablecoins now runs detectDepegEvents and confirmPendingDepegs sequentially each cycle
- Secondary agreement uses CoinGecko and/or DEX with a 50% threshold bar

Details

v2.1 Feb 20, 2026 Reconstructed

Four-year peg-score lookback window

Peg scoring moved to an explicit rolling 4-year horizon instead of unbounded historical span.

- computePegScoreWithWindow introduced a 4-year tracking cap
- Boundary logic was later corrected in v3.2 for firstSeen direction
- Detail-page peg scores became explicitly lookback-bounded

Details

v2.0 Feb 20, 2026 Reconstructed

Peg-score severity rebalance + spread penalty

Peg score shifted to stronger magnitude sensitivity and penalized erratic event-size variance.

- Severity penalty changed from sqrt-based to linear (peakBps/100 scaling)
- Composite became: 0.5*pegPct + 0.5*severity - activePenalty - spreadPenalty
- Added spreadPenalty from event-magnitude standard deviation (cap 15)

Details

v1.2 Feb 20, 2026 Reconstructed

Non-USD thresholding + ongoing false-positive control

Depeg detection adopted peg-type-aware thresholds and began actively retiring stale false-positive open events.

- Non-USD depeg threshold raised to 150 bps (USD remained 100 bps)
- Ongoing events auto-close after sustained DEX disagreement (30m+, >=$1M TVL)
- Cleanup migration removed legacy non-USD events below 150 bps

Details

v1.1 Feb 18, 2026 Reconstructed

Early lifecycle hardening + active penalty floor

Early stability pass reduced under-penalization and fixed event-time accounting/state-lifecycle edge cases.

- Added active-depeg penalty to peg score, then introduced a minimum floor
- Detection now closes orphan open events and tracks open state by event ID
- Merged overlapping depeg intervals to prevent pegPct double-counting

Details

v1.0 Feb 18, 2026 Reconstructed

Initial Depeg Tracker scoring + live event detection

First operational release of depeg scoring and event detection primitives.

- Launched computePegScore baseline (peg time + severity blend)
- Added duplicate-open-event merge and new-event DEX disagreement suppression
- Introduced detectDepegEvents cron pipeline with live open/close/update logic

Details

WHITE PAPER Download Depeg Tracker + DEWS Changelog v6.0 (PDF)

Cite this page

Accessed 2026-05-17. `urn:pharos:methodology:dews@v6.0`

BibTeX | APA 7 | Chicago | URL + URN

```
@misc{methodology_dews_v6-0,
  author       = {Pharos Watch},
  title        = {Depeg Tracker + DEWS Changelog},
  year         = {2026},
  month        = {may},
  howpublished = {\url{https://pharos.watch/methodology/depeg-changelog/}},
  version      = {v6.0},
  note         = {Accessed: 2026-05-17; urn:pharos:methodology:dews@v6.0},
  urldate      = {2026-05-17}
}
```